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Sequential Training of Windowed PINNs for the Lorenz-1960 ODE System

Ikramul Hasan Moral, Md. Abu Bakar, Samiur Rahman Omlan,
Md. Touhidul Islam, Fariha Islam, Dr. Muhammad Nomani Kabir
Department of Computer Science and Engineering, United International University, Dhaka, Bangladesh
imoral223489@bscse.uiu.ac.bd, mbakar223200@bscse.uiu.ac.bd, somlan223195@bscse.uiu.ac.bd,
mislam223435@bscse.uiu.ac.bd, fislam223431@bscse.uiu.ac.bd, kabir@cse.uiu.ac.bd

***Abstract*—We study a physics-informed neural network (PINN) approximation of a single nearly closed orbit of the Lorenz three-state system on $t \in [0, 13.26446]$. We trained a single network on the full interval that produces a combined state root-mean-square error (RMSE) of 2.237. To address this long horizon problem, we divide the interval into 27 windows. Each window has a separate network and an initial state that is fixed to the predicted endpoint of the previous window. In each window, Adam reduces the residual in four stages. In each stage, we use causal weights and causal weighted loss to reduce the loss of all earlier points. Then, L-BFGS minimizes the loss using the unweighted residual mean. A numerical reference is used to monitor the training but is not used for optimization or stopping. Across five seeds, the windowed model gives a median combined RMSE of 8.13×10^{-5} on 13,265 evaluation times. Also a dense-grid residual mean square of 1.87×10^{-7}. The full-interval and windowed runs also vary in number of parameters, sampling and optimizer schedule, so we describe them as different configurations. Ablations show that separate window networks is needed and warm start improves accuracy and reduced training steps.**

***Index Terms*—physics-informed neural network, Lorenz system, causal training, time windows, ordinary differential equation, windowed PINN**

I. Introduction

The Lorenz-system model is a three-state, nonlinear ordinary differential equation (ODE) that was obtained from a reduced flow model [1]. It has an initial-value problem which provides a simple test of the convergence of a learned continuous-time function from a specified trajectory over a long interval. A PINN can't replace the numerical solvers for problem like this, but PINNs are still usefull to recover unknown coefficients [2] and those use cases still needs to train well for a long time interval. Long horizon training is a known failure case of PINNs [3], [4]. That is the reason we choose this system , its reference solution is known to high accuracy. So a reference trajectory is provided by numerical integration. A PINN is trained with the residual of the ODE instead of the solution samples [2]. Previous works do not suggest training a long horizon problem with a traditional PINN [3]. It couples early and late times, but the state at a later time is determined by the previous trajectory. To overcome this ordering, Wang *et al.* suggested time-dependent residual weights for PINN training [3]. An initial condition can also be met by construction using a hard trial solution [5].

We use both concepts. We split the time period into 27 equal windows and use a separate network for every window. The initial state of each window is fixed to the last collocation point predicted by the previous window. The parameters of a trained window are copied to the next trainable window. We report one implementation and its saved result on $[0, 13.26446]$, which is a nearly closed orbit for the chosen initial state. The actual training goal and window handoff are given and compared to a saved single-network training of the same initial-value problem. The comparison is descriptive because multiple configuration options are changing simultaneously. Our contributions are a windowed PINN for one nearly closed Lorenz orbit, where 27 networks are trained in sequence with a hard endpoint handoff, warm start and staged causal weights (Section III); a multi-seed run with median combined RMSE of 8.13×10^{-5} against DOP853 (Section V); and ablation and failure cases showing that separate window networks and warm start are needed, while the gain from causal weighting is not significant in all runs (Section V-C).

II. Problem and Background

In this section we describe the Lorenz equations and the initial value problem that we want the model to learn. Then we explain the existing methods or components that we used.

A. Lorenz Equations

Our main object is a three-state nonlinear ordinary differential equation (ODE) that we need to solve. The system is non-chaotic and periodic and the time range approximates the period of a single loop. Here $x(t)$, $y(t)$ and $z(t)$ are the three states, and κ and ℓ are the physical wavenumbers. We use k only for the window index, which comes later. The Lorenz system general form [1] is:

$$\begin{aligned} \frac{dx}{dt} &= \kappa\ell \left(\frac{1}{\kappa^2 + \ell^2} - \frac{1}{\kappa^2} \right) yz, \\ \frac{dy}{dt} &= \kappa\ell \left(\frac{1}{\ell^2} - \frac{1}{\kappa^2 + \ell^2} \right) xz, \\ \frac{dz}{dt} &= \frac{\kappa\ell}{2} \left(\frac{1}{\kappa^2} - \frac{1}{\ell^2} \right) xy. \end{aligned} \tag{1}$$

Suppose the state vector is given by $u(t) = [x(t), y(t), z(t)]^\mathsf{T}$. We set $\kappa = 2$ and $\ell = 1$ in (1) and get the three coefficients.

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$$a_y = 2(1 - 1/5) = 8/5 = 1.6,$$
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Using these coefficient values, we get the reduced system we used in this experiment.

$$\dot{x} = -0.1\,yz, \qquad \dot{y} = 1.6\,xz, \qquad \dot{z} = -0.75\,xy, \quad (2)$$

If we collect the three right-hand sides, we get the vector form $du/dt = f(u)$. Here f is the known ODE vector field.

$$f(u) = [a_x yz,\, a_y xz,\, a_z xy]^\mathsf{T}, \qquad f : \mathbb{R}^3 \to \mathbb{R}^3.$$

The input of f is a state vector u with three components and the output is the rate of change of x, y and z in the same order. While training, we use this same function on the predicted state $\hat{u}_k(t)$.

For our experiment we used the coefficients $a = [a_x, a_y, a_z]^\mathsf{T} = [-0.1, 1.6, -0.75]^\mathsf{T}$. The system starts from an initial state [6] of $u(0) = [0.5, 0.75, 1.0]^\mathsf{T}$ until a final time of $T = 13.26446$. At this endpoint, the reference trajectory nearly returns to its starting state. It is very similar but not exactly closed so we call this interval a nearly closed orbit and not an exact period.

B. Related Work

In this study we are making an experimental investigation of a series of well known concepts. The approach adopted is a PINN that embeds the physics residual into the training objective [2]. We use a hard trial-function design [5] to fix our starting state exactly at the point we want. Learning long trajectories is a well-known challenge due to the ordering of the time axis so we take inspiration from the causal residual weighting technique [3]. Lastly segmenting the timeline is based on ideas of causal sweeping and temporal decomposition [7]. Our implementation is an investigation of the performance of the individual components in this nearly closed orbit of the Lorenz system.

Several works explain why PINN training fails. Krishnapriyan *et al.* showed that PINNs can fail on simple problems and that learning the solution in short time segments helps [4]. Wang *et al.* linked the failure to imbalanced gradients and to the neural tangent kernel of the loss terms [8], [9]. For dynamical systems, fixed points can trap training in a wrong solution [10], and the loss landscape is ill-conditioned, so Adam followed by L-BFGS trains better than Adam alone [11]. Mattey and Ghosh also trained time segments in sequence, but with one network that is retrained on each new segment [12].

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This section explains the implementation of windowed PINN. It also contains the model construction, physics enforcement, causal weights, window segmentation, its handoff and the sequence in how the optimizer works for this particular case.

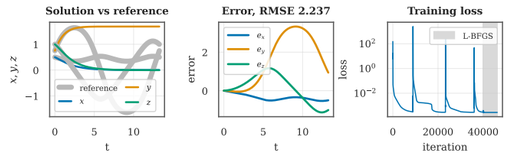


Fig. 1. Single network on the full interval

A. Network Structure

We used several modeling strategies to approximate the solution for Lorenz system trajectory over $[0, T]$, where $T = 13.26446$ is one nearly closed loop for the chosen initial state. We first tried one network to learn the entire trajectory at once. As it got stuck in a plateau, we tried to split the time interval in equal segments.

We chose a window width close to 0.5. In the first experiment, we used one network with 27 segmented time period to train the model sequentially. We noticed that, whenever the window is moved to the new one, the previous learning is lost or degraded. This results in a significant increase in error with RMSE of 0.350 to 0.648 over three seeds (Table I). To address these issues, we implemented WPINN (Windowed PINN). We split the total time $[0, T]$ equally into 27 windows and gave each window its own network. Here $k = 0, \ldots, 26$ is the index of the 27 windows and $t_0, \ldots, t_{27}$ are the 28 boundary times. Each window W_k has width $\Delta t = T/27$ and its own network parameters θ_k.

$$t_k = kT/27, \quad k = 0, \ldots, 27, \quad W_k = [t_k, t_{k+1}],$$
$$\Delta t = T/27 \approx 0.4912763, \quad N_k(\,t; \theta_k) : W_k \to \mathbb{R}^3.$$

Each window has its own network $N_k(t; \theta_k)$ with 1 input layer (t), 4 hidden layers of 60 neurons with tanh unit, and an output layer with 3 outputs. The output is not the state (x, y, z) directly, it is a correction vector. We get the predicted state $\hat{u}_k(t)$ from the trial solution in (3). With biases, the input layer has 120 parameters, the three hidden-to-hidden layers have 3×3660 and the output layer has 183. The total counts all 27 networks.

$$P_{\text{window}} = 120 + 3(3660) + 183 = 11{,}283, \; P_{\text{total}} = 304{,}641.$$

For the model parameter initialization we used xavier-uniform weights and zero biases. The 27 networks do not share parameters, but neighboring windows share a boundary. The predicted endpoint of a completed window is used as the initial state of the next window.

B. Window Handoff

As the Lorenz system is an initial value problem, we enforce the initial state of each window through a trial solution. Here t_k is the starting time of window k, $u_{0,k} \in \mathbb{R}^3$ is its fixed starting state and $N_k(t; \theta_k) \in \mathbb{R}^3$ is the network raw output.

$$\hat{u}_k(t) = u_{0,k} + (t - t_k) N_k(t; \theta_k), \quad t \in W_k. \quad (3)$$

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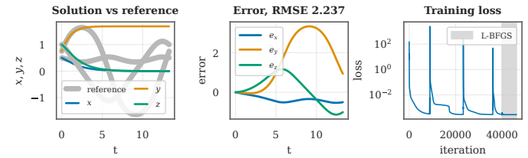


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Fig. 1 shows the saved full-interval result.

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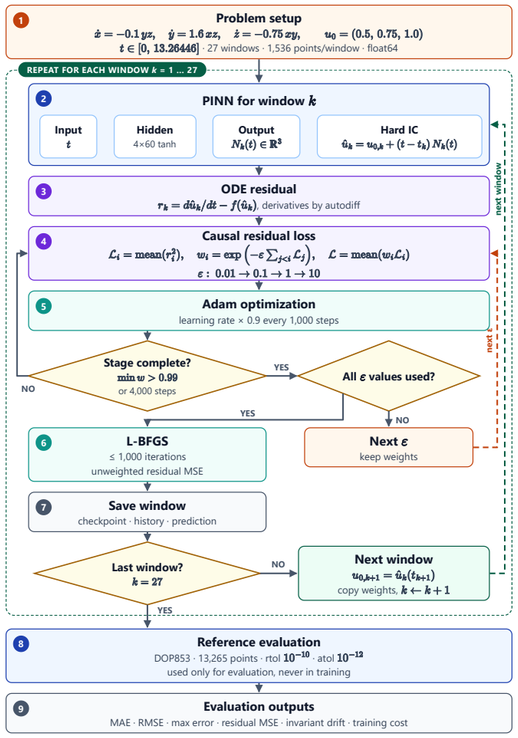


Fig. 2. Implementation pipeline (windows numbered 1-27)

At $t = t_k$, the second term is zero, so $\hat{u}_k(t_k) = u_{0,k}$ regardless of the network parameters [5]. The first window uses the given initial state $u_{0,0} = u(0)$, and every later window uses the predicted endpoint of the last window as its initial state.

$$u_{0,k+1} = \hat{u}_k(t_{k+1}), \qquad k = 0, \ldots, 25. \tag{4}$$

The endpoint transfer in (4) and the trial solution make the predicted values continuous at the window boundaries. But the derivatives can still change suddenly at the boundaries. Window $k+1$ starts from a copy of the weights of window k, $\theta_{k+1} \leftarrow \theta_k$ for $k = 0, \ldots, 25$, and then trains its own parameters. As the starting point of the later windows is predicted, it can carry error from earlier windows.

C. Training Loss

To train the network of each window, we split the time range into $n_k = 1536$ uniform collocation points $t_{k,i}$, $i = 1, \ldots, n_k$, including both endpoints.

$$t_{k,i} = t_k + \frac{i-1}{n_k - 1}(t_{k+1} - t_k),$$
$$n_{\text{unique}} = 27(1536 - 1) + 1 = 41{,}446, \quad 27 \cdot 1536 = 41{,}472.$$

Two windows next to each other share their boundary time. So there are 41,446 unique collocation points in total, but 41,472 collocation points are used in training. On the global grid, each boundary point belongs to the next window, and we also add it when we train the current window. The last window has the final endpoint. While training, all the points in a single window are passed to the network at once which gives an output of shape $(1536, 3)$. Then we apply the trial solution to enforce the initial value. PyTorch automatic differentiation gives the rate of change of x, y and z with respect to time for all the collocation points. As $u_{0,k}$ is fixed while training, the product rule gives the derivative of (3):

$$\frac{d\hat{u}_k}{dt} = N_k + (t - t_k)\frac{\partial N_k}{\partial t}.$$

The ODE residual function compares this derivative with $f(\hat{u}_k(t))$, which is the original ODE right-hand side at the predicted state. Both terms are vectors with three components, and their difference is the residual $r_k(t)$.

$$r_k(t) = \frac{d\hat{u}_k(t)}{dt} - f\big(\hat{u}_k(t)\big) \in \mathbb{R}^3. \tag{5}$$

So the three components are $r_{k,x} = d\hat{x}_k/dt + 0.1\,\hat{y}_k\hat{z}_k$, $r_{k,y} = d\hat{y}_k/dt - 1.6\,\hat{x}_k\hat{z}_k$ and $r_{k,z} = d\hat{z}_k/dt + 0.75\,\hat{x}_k\hat{y}_k$.

For a batch of n_k times, each row is one collocation point and the three columns are x, y and z. Here $\mathbf{N}_k$, $\hat{\mathbf{u}}_k$, $\dot{\hat{\mathbf{u}}}_k$, $\mathbf{f}_k$ and $\mathbf{r}_k$ mean the network outputs, trial states, time derivatives, ODE right-hand sides and residuals. Each of them is in $\mathbb{R}^{n_k \times 3}$, and the time batch $\mathbf{t}_k$ is in $\mathbb{R}^{n_k \times 1}$.

For each ordered collocation time $t_{k,i}$, we take the mean of the three squared residual components. This gives one non-negative loss value $L_{k,i}$ for each point instead of three components:

$$L_{k,i} = \tfrac{1}{3}\,\|r_k(t_{k,i})\|_2^2 = \tfrac{1}{3}\big(r_{k,x}^2 + r_{k,y}^2 + r_{k,z}^2\big)(t_{k,i}). \tag{6}$$

For a long horizon time dependent ODE system, residual loss alone may not be sufficient. That is why we used causal weights and causal weighted loss. The causal weight at a time depends on the accumulated residual loss at earlier times [3]. Here $S_{k,i}$ is the sum of the point losses at all earlier times of the same window. The first point has no earlier point, so $S_{k,1} = 0$ and its causal weight is one.

$$w_{k,i} = \exp\left(-\epsilon S_{k,i}\right) = \exp\left(-\epsilon \sum_{j=1}^{i-1} L_{k,j}\right). \tag{7}$$

So $0 < w_{k,i} \leq 1$. Then we multiply each point loss with its causal weight and take the mean over all n_k points. This gives one loss value for the window in the current ϵ stage:

$$\mathcal{L}_{k,\epsilon} = \frac{1}{n_k}\sum_{i=1}^{n_k} w_{k,i} L_{k,i} \in \mathbb{R}_{\geq 0}. \tag{8}$$

Here k is the window, i is the current point and $j < i$ means the earlier points of i. $\mathbf{L}_k, \mathbf{w}_k \in \mathbb{R}^{n_k}$ have one loss and one weight for each point. ϵ in the equation means how strongly the earlier losses reduce the later weights. We used four ϵ stages:

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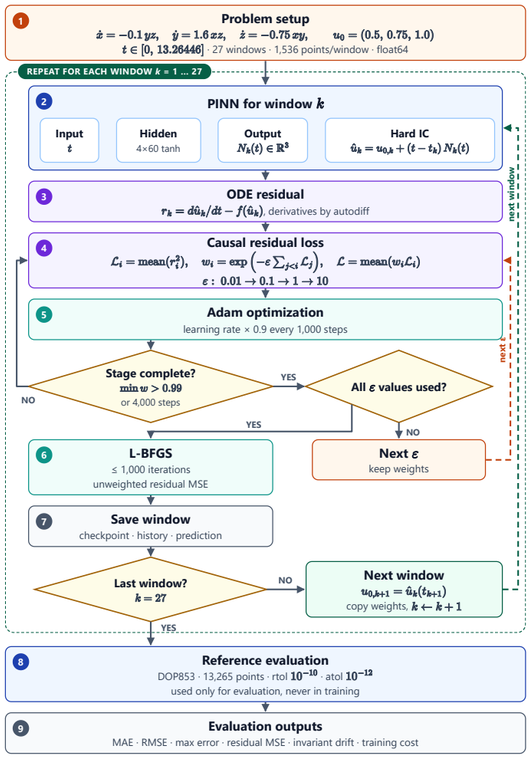


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$$\frac{d\hat{u}_k}{dt} = N_k + (t - t_k)\frac{\partial N_k}{\partial t}.$$

The ODE residual function compares this derivative with $f(\hat{u}_k(t))$, which is the original ODE right-hand side at the predicted state. Both terms are vectors with three components, and their difference is the residual $r_k(t)$.

$$r_k(t) = \frac{d\hat{u}_k(t)}{dt} - f\big(\hat{u}_k(t)\big) \in \mathbb{R}^3. \tag{5}$$

So the three components are $r_{k,x} = d\hat{x}_k/dt + 0.1\,\hat{y}_k\hat{z}_k$, $r_{k,y} = d\hat{y}_k/dt - 1.6\,\hat{x}_k\hat{z}_k$ and $r_{k,z} = d\hat{z}_k/dt + 0.75\,\hat{x}_k\hat{y}_k$.

For a batch of n_k times, each row is one collocation point and the three columns are x, y and z. Here $\mathbf{N}_k$, $\hat{\mathbf{u}}_k$, $\dot{\hat{\mathbf{u}}}_k$, $\mathbf{f}_k$ and $\mathbf{r}_k$ mean the network outputs, trial states, time derivatives, ODE right-hand sides and residuals. Each of them is in $\mathbb{R}^{n_k \times 3}$, and the time batch $\mathbf{t}_k$ is in $\mathbb{R}^{n_k \times 1}$.

For each ordered collocation time $t_{k,i}$, we take the mean of the three squared residual components. This gives one non-negative loss value $L_{k,i}$ for each point instead of three components:

$$L_{k,i} = \tfrac{1}{3}\,\|r_k(t_{k,i})\|_2^2 = \tfrac{1}{3}\big(r_{k,x}^2 + r_{k,y}^2 + r_{k,z}^2\big)(t_{k,i}). \tag{6}$$

For a long horizon time dependent ODE system, residual loss alone may not be sufficient. That is why we used causal weights and causal weighted loss. The causal weight at a time depends on the accumulated residual loss at earlier times [3]. Here $S_{k,i}$ is the sum of the point losses at all earlier times of the same window. The first point has no earlier point, so $S_{k,1} = 0$ and its causal weight is one.

$$w_{k,i} = \exp\left(-\epsilon S_{k,i}\right) = \exp\left(-\epsilon \sum_{j=1}^{i-1} L_{k,j}\right). \tag{7}$$

So $0 < w_{k,i} \leq 1$. Then we multiply each point loss with its causal weight and take the mean over all n_k points. This gives one loss value for the window in the current ϵ stage:

$$\mathcal{L}_{k,\epsilon} = \frac{1}{n_k}\sum_{i=1}^{n_k} w_{k,i} L_{k,i} \in \mathbb{R}_{\geq 0}. \tag{8}$$

Here k is the window, i is the current point and $j < i$ means the earlier points of i. $\mathbf{L}_k, \mathbf{w}_k \in \mathbb{R}^{n_k}$ have one loss and one weight for each point. ϵ in the equation means how strongly the earlier losses reduce the later weights. We used four ϵ stages:

0.01, 0.1, 1, and 10. When the earlier losses go down, the later weights go up towards one. The weights are recomputed during training, and they are fixed while the Adam optimizer calculates the gradient. The causal weights also depend on the number and spacing of the collocation points [3].

Only using the ODE residual is not enough for a long horizon system like the Lorenz system. Especially for the windowed algorithm, early times should be fitted before late times, as the errors accumulate and backtracking is not possible. Therefore, later states depend on the earlier states. If earlier error is reduced, possibly the later error can be reduced as well. Equations (6)-(8) do this with the point losses and the causal weights. The next subsection explains the stopping rules of the stages and the unweighted L-BFGS refinement steps.

D. Training Procedure

The causal loss defined above is minimized separately in each window. Figure 2 summarizes the sequence across the windows. Algorithm 1 gives the training steps. In each window an Adam optimizer and a StepLR scheduler are created. The learning rate starts at 10^{-3} and it is multiplied by 0.9 for every 1,000 updates across all four ϵ stages of the window. There is a 4,000 update limit for each stage. After every Adam update the code recalculates the causal weights at all collocation points of the window. If the minimum causal weight of all 1536 points of the window is above 0.99 after an update, then it moves to the next ϵ stage. If it is below 0.99 and it reaches the update limit then it moves to the next stage and records failure.

When all four stages are completed, L-BFGS tries to minimize the loss using unweighted residual mean up to 1,000 iterations. For L-BFGS, we remove the causal weights and use the plain mean of the point losses.

$$\mathcal{L}_k^{\mathrm{raw}} = \frac{1}{n_k}\sum_i L_{k,i} = \frac{1}{3n_k}\sum_i \|r_k(t_{k,i})\|_2^2.$$

To keep track of training, we save a snapshot every 50 updates. After each window the trained parameters, predicted endpoint are stored. The same parameters initialize the next window. Reference grid values are only used for evaluation. After training, the model is evaluated on the collocation points and the evaluation grid to get the error, residual and other measurements.

IV. Experimental Setup

This section explains the experimental setup for the saved windowed run. It contains the configuration and reference information, the evaluation measures and the separate full interval comparison.

A. Reference Solution

Table II shows the parameters used for the reported windowed run. We use SciPy's DOP853 [13] as the numerical reference, a relative tolerance of 10^{-10} and an absolute tolerance of 10^{-12}. It is sampled at 13,265 equally-spaced time points on a separate evaluation grid.

Algorithm 1 Sequential window training

Require: Ordered windows $W_0, \ldots, W_{26}$ and their uniform time points

Set $u_{0,0} \leftarrow [0.5, 0.75, 1.0]^{\mathsf{T}}$
for $k = 0, \ldots, 26$ **do**
  if $k > 0$ **then**
    Set $u_{0,k} \leftarrow \hat{u}_{k-1}(t_k)$
    Initialize $\theta_k \leftarrow \theta_{k-1}$
  end if
  Add t_{k+1} to the residual points if $k < 26$
  for $\epsilon \in (0.01, 0.1, 1, 10)$ **do**
    for at most 4,000 Adam steps **do**
      Use ordered times $t_{k,i}$, $i = 1, \ldots, n_k$
      Form $\hat{u}_k$ by (3); get $d\hat{u}_k/dt$ by autograd
      Form r_k by (5) and $L_{k,i}$ by (6)
      Compute $w_{k,i}$ by (7); detach weights
      Backpropagate (8); Adam then StepLR
      if verified post-update $\min_i w_{k,i} > 0.99$ **then**
        break
      end if
    end for
    Record whether the threshold was met or capped
  end for
  Run L-BFGS on the unweighted residual loss
  Save window model, endpoint state, and config record
end for

B. Evaluation

At each evaluation time t_j, $j = 1, \ldots, n_{\mathrm{eval}}$, we subtract the reference state from the predicted state and get the error vector $e(t_j) \in \mathbb{R}^3$. This error is not the same as the ODE residual.

$$e(t_j) = \hat{u}(t_j) - u_{\mathrm{ref}}(t_j) = [e_x, e_y, e_z]^{\mathsf{T}}(t_j).$$

For the combined RMSE, we take the mean of the squared error norms over all evaluation times and then the square root. It is also equal to the square root of the sum of the three squared component RMSEs.

$$\mathrm{RMSE}_2 = \left(\frac{1}{n_{\mathrm{eval}}}\sum_{j=1}^{n_{\mathrm{eval}}} \|e(t_j)\|_2^2\right)^{1/2}. \tag{9}$$

For each state $c \in \{x, y, z\}$ we also report MAE, RMSE and the maximum absolute error. E_{mean} and $E_{\max}$ are the mean and the maximum of the error norm.

The dense-grid residual MSE is the average of the three residual components squared, taken over the times of the evaluation. So it has a factor of $1/(3n_{\mathrm{eval}})$, but the combined state RMSE is not divided by three.

$$\mathrm{MSE}_{r,\mathrm{eval}} = \frac{1}{3n_{\mathrm{eval}}}\sum_{j=1}^{n_{\mathrm{eval}}} \|r(t_j)\|_2^2.$$

The comparison run uses the same network depth and width on the entire interval $[0, T]$, and we compare their final outputs.

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When all four stages are completed, limited-memory Broyden–Fletcher–Goldfarb–Shanno (L-BFGS) tries to minimize the loss using unweighted residual mean up to 1,000 iterations. For L-BFGS, we remove the causal weights and use the plain mean of the point losses.

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To keep track of training, we save a snapshot every 50 updates. After each window the trained parameters, predicted endpoint are stored. The same parameters initialize the next window. Reference grid values are only used for evaluation. After training, the model is evaluated on the collocation points and the evaluation grid to get the error, residual and other measurements.

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      if verified post-update $\min_i w_{k,i} > 0.99$ **then**
        break
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    end for
    Record whether the threshold was met or capped
  end for
  Run L-BFGS on the unweighted residual loss
  Save window model, endpoint state, and config record
end for

Fig. 3. Sequential window training

B. Evaluation

At each evaluation time t_j, $j = 1, \ldots, n_{\mathrm{eval}}$, we subtract the reference state from the predicted state and get the error vector $e(t_j) \in \mathbb{R}^3$. This error is not the same as the ODE residual.

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For each state $c \in \{x, y, z\}$ we also report mean absolute error (MAE), RMSE and the maximum absolute error. E_{mean} and $E_{\max}$ are the mean and the maximum of the error norm.

The dense-grid residual mean square error (MSE) is the average of the three residual components squared, taken over the times of the evaluation. So it has a factor of $1/(3n_{\mathrm{eval}})$, but the combined state RMSE is not divided by three.

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The comparison run uses the same network depth and width on the entire interval $[0, T]$, and we compare their final outputs.

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TABLE I
ABLATION RESULTS (MEDIANS OVER SEEDS)

Configuration	Seeds	Median RMSE	RMSE range	Residual MSE
WPINN	5	8.13×10^{-5}	6.97×10^{-5}–2.60×10^{-4}	1.87×10^{-7}
No causal	3	1.89×10^{-4}	1.75×10^{-4}–5.16×10^{-4}	8.81×10^{-7}
No warm start	3	2.85×10^{-4}	1.76×10^{-4}–2.88×10^{-4}	8.35×10^{-7}
Shared network	3	0.486	0.350–0.648	0.965
Causal single	1	0.275	–	5.23×10^{-6}
Full interval	1	2.237	–	2.80×10^{-4}

TABLE II
WINDOWED-RUN CONFIGURATION

Setting	Value
Interval; windows	$[0, 13.26446]$; 27 equal windows
Network per window	1-60-60-60-60-3, tanh
Parameters	11,283/window; 304,641 total
Initial condition; precision	$[0.5, 0.75, 1.0]^{\mathsf{T}}$; float64
Uniform collocation grid	41,446 unique points
Adam ϵ stages	0.01, 0.1, 1, 10
Stage threshold; steps/stage	$\min_{1\le i\le n_k} w_{k,i} > 0.99$; 4,000
Adam rate; scheduler	10^{-3}; $\times0.9$ per 1,000 steps
L-BFGS budget	at most 1,000 iterations/window
Warm start; seed	Initialize $\theta_k \leftarrow \theta_{k-1}$; 0
Reference grid	13,265 uniform times

TABLE III
WINDOWED-RUN ERROR (SEED 0)

State	MAE	RMSE	Maximum
x	1.89×10^{-5}	2.74×10^{-5}	1.13×10^{-4}
y	2.94×10^{-5}	4.96×10^{-5}	2.88×10^{-4}
z	2.78×10^{-5}	4.05×10^{-5}	1.30×10^{-4}
State norm	5.12×10^{-5}	6.97×10^{-5}	3.09×10^{-4}

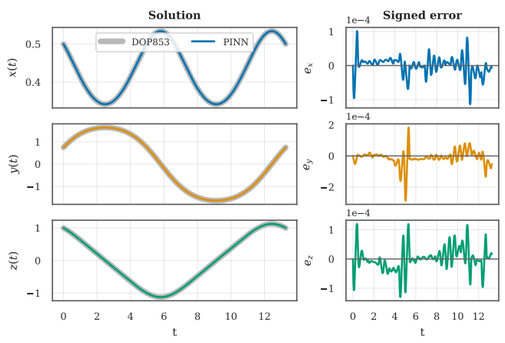


Fig. 3. Windowed PINN vs. DOP853 (seed 0)

C. Ablation Protocol

Table I shows the ablation and failure-case runs. All runs use the same initial state, coefficients, float64 precision, interval and DOP853 reference grid. The WPINN row is our windowed PINN from Section III, and we ran it with seeds 0 - 4. Seed 0 is the run we reported in Tables III and IV. For the ablations we used seeds 1-3. In the no-causal case we removed the causal weights and trained each window for 1,300 Adam steps (35,100 in total). This is close to the WPINN, which used 34,768 steps on average. In the no-warm-start case we kept the WPINN schedule but did not copy the parameters of the previous window. The shared-network case uses the no-causal schedule, but one network is trained for all 27 windows one after another. In the causal single-network case we used causal weights on one network for the full interval, with the same grid and budget as the full-interval run, 39,793 LHS points, 40,000 Adam steps and at most 5,000 L-BFGS iterations. Some runs were on CPU and some on GPU, so we compare Adam steps.

D. Failure Cases

We used two failure cases. The shared-network case checks if one network can keep what it learned in the earlier windows. The causal single-network case checks if causal weighting alone can fix the error of the full-interval network. We judge a run only by its final error against DOP853. All the runs completed training, so a large error here means a poor prediction and not a stopped job. When a stage reaches the 4,000-step cap we record it, but this alone does not mean the trajectory failed. We did not use any success cutoff to pick runs, and Table I shows every completed seed.

V. Results

In this section we report the accuracy and physics checks of the windowed run against DOP853. Then we compare it with the full-interval run and we also show the ablation and failure results.

A. Prediction Accuracy

We compare the windowed PINN with an extremely accurate numerical reference which is SciPy's DOP853 and RK4. As shown in Table III, our model achieved a median combined RMSE of 8.13×10^{-5} across the 13,265 uniform times in our evaluation grid and the best RMSE across the seeds is 6.97×10^{-5}. The highest state error norm was 3.09×10^{-4}. Figure 3 plots these predictions against the reference over time. The trajectories overlap completely, but the error shows small peaks, particularly at $t \simeq 5$. In the state-norm row, the MAE-column entry is E_{mean}, the RMSE entry is RMSE_2 from (9), and the maximum entry is $E_{\max}$.

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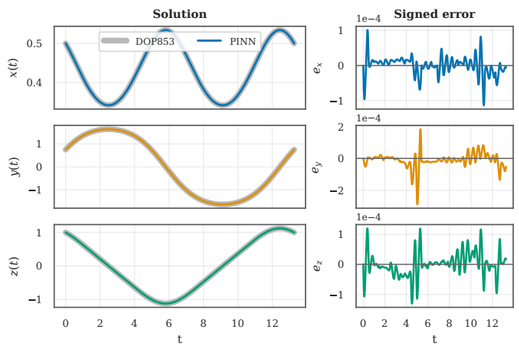


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TABLE IV
FULL-INTERVAL AND WINDOWED RUNS

Measure	One network	27 windows
Trainable parameters	11,283	304,641
Training points	39,793 LHS	41,446 uniform
Combined state RMSE	2.237	6.97×10^{-5}
Maximum state-error norm	3.346	3.09×10^{-4}
Collocation residual MSE	2.80×10^{-4}	1.87×10^{-7}
Recorded Adam steps	40,000	34,765

We also measured how closely the model obeyed the underlying equations. The MSE of the ODE residual was very small on the evaluation grid at 1.872×10^{-7}. The residual MSE of the windowed run on the collocation grid is nearly the same (1.874×10^{-7}). The system (2) conserves every quadratic form $\alpha x^2 + \beta y^2 + \gamma z^2$ with $[\alpha, \beta, \gamma]^{\mathsf{T}}$ orthogonal to a. The maximum relative drift of its two independent invariants is below 4×10^{-10} for DOP853 and below 2.6×10^{-4} for our PINN.

Lastly, the hard trial solution ensures that the values of the state are exactly continuous at the inner 26 interfaces connecting the windows.

B. Run Comparison

The differences with respect to the two saved configurations are shown in Table IV. The RMSE value of the entire interval network is 2.237,the network satisfies the ODE closely but on the wrong trajectory, consistent with [10]. This RMSE is much larger than the windowed run network's RMSE value of 6.97×10^{-5}. It is a result for the recorded settings of this initial value problem. This is a comparison of two saved configurations, which differ in causal weighting [3], windowing [7] and warm starts as well as the collocation grid, the Adam schedule and the L-BFGS budget. The full-interval row of Table I is this same run. In particular the model with the window is much more complicated with 27 times as many parameters and a different collocation grid.

The 27 run windows had 34,765 Adam updates and 33,143 L-BFGS closure evaluations in total. Closure evaluations are function calls made by L-BFGS, not iterations. Only two of the 108 causal stages ended at the 4,000-step cap without meeting the 0.99 weight threshold.

C. Ablation and Failure Results

Table I shows the ablation results. The median RMSE of the WPINN is 8.13×10^{-5}, and three of the five seeds are below 10^{-4}. Without causal weights the median is about 2.3 times larger, but the difference is not significant (Mann-Whitney $p = 0.196$), and for seed 2 the no-causal run has lower RMSE than the WPINN. Without warm start, all three seeds have larger RMSE than the WPINN with the same seed. The mean Adam steps also rise to 102,989, and 17-19 of the 108 stages reach the cap, while the WPINN has only 1-2. When one network is reused for all windows, the RMSE is between 0.350 and 0.648. With separate networks and the same no-causal schedule it is between 1.75×10^{-4} and 5.16×10^{-4}. This matches what we noticed in Section III, that the previous learning is lost or degraded when one network moves to a new window. The causal single network gives 0.275 in its one run. This is lower than the full-interval run but still far from the windowed runs. These runs support the combination of windows, warm start and causal weights that we tested for this orbit. But with three to five seeds, they do not show that causal weighting always improves the accuracy.

VI. Conclusion

Our 27-window PINN passes every check on the nearly closed Lorenz-system interval $[0, 13.26446]$. The seed-0 run had a combined state RMSE of 6.97×10^{-5} in comparison to DOP853, and the median over five seeds is 8.13×10^{-5}. A separate network trained on the full interval recorded an RMSE of 2.237. This research applies to only one initial state and one coefficient set. Other initial states and other coefficient sets may result in different behavior. Future research will focus on matched configurations and initial states and generalize to problems with specified boundary conditions at each end.

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TABLE IV
FULL-INTERVAL AND WINDOWED RUNS

Measure	One network	27 windows
Trainable parameters	11,283	304,641
Training points	39,793 LHS	41,446 uniform
Combined state RMSE	2.237	6.97×10^{-5}
Maximum state-error norm	3.346	3.09×10^{-4}
Collocation residual MSE	2.80×10^{-4}	1.87×10^{-7}
Recorded Adam steps	40,000	34,765

LHS: Latin hypercube sampling.

We also measured how closely the model obeyed the underlying equations. The MSE of the ODE residual was very small on the evaluation grid at 1.872×10^{-7}. The residual MSE of the windowed run on the collocation grid is nearly the same (1.874×10^{-7}). The system (2) conserves every quadratic form $\alpha x^2 + \beta y^2 + \gamma z^2$ with $[\alpha, \beta, \gamma]^{\mathsf{T}}$ orthogonal to a. The maximum relative drift of its two independent invariants is below 4×10^{-10} for DOP853 and below 2.6×10^{-4} for our PINN.

Lastly, the hard trial solution ensures that the values of the state are exactly continuous at the inner 26 interfaces connecting the windows.

B. Run Comparison

The differences with respect to the two saved configurations are shown in Table IV. The RMSE value of the entire interval network is 2.237,the network satisfies the ODE closely but on the wrong trajectory, consistent with [10]. This RMSE is much larger than the windowed run network's RMSE value of 6.97×10^{-5}. It is a result for the recorded settings of this initial value problem. This is a comparison of two saved configurations, which differ in causal weighting [3], windowing [7] and warm starts as well as the collocation grid, the Adam schedule and the L-BFGS budget. The full-interval row of Table I is this same run. In particular the model with the window is much more complicated with 27 times as many parameters and a different collocation grid.

The 27 run windows had 34,765 Adam updates and 33,143 L-BFGS closure evaluations in total. Closure evaluations are function calls made by L-BFGS, not iterations. Only two of the 108 causal stages ended at the 4,000-step cap without meeting the 0.99 weight threshold.

C. Ablation and Failure Results

Table I shows the ablation results. The median RMSE of the WPINN is 8.13×10^{-5}, and three of the five seeds are below 10^{-4}. Without causal weights the median is about 2.3 times larger, but the difference is not significant (Mann-Whitney $p = 0.196$), and for seed 2 the no-causal run has lower RMSE than the WPINN. Without warm start, all three seeds have larger RMSE than the WPINN with the same seed. The mean Adam steps also rise to 102,989, and 17-19 of the 108 stages reach the cap, while the WPINN has only 1-2. When one network is reused for all windows, the RMSE is between 0.350 and 0.648. With separate networks and the same no-causal schedule it is between 1.75×10^{-4} and 5.16×10^{-4}. This matches what we noticed in Section III, that the previous learning is lost or degraded when one network moves to a new window. The causal single network gives 0.275 in its one run. This is lower than the full-interval run but still far from the windowed runs. These runs support the combination of windows, warm start and causal weights that we tested for this orbit. But with three to five seeds, they do not show that causal weighting always improves the accuracy.

VI. Conclusion

Our 27-window PINN passes every check on the nearly closed Lorenz-system interval $[0, 13.26446]$. The seed-0 run had a combined state RMSE of 6.97×10^{-5} in comparison to DOP853, and the median over five seeds is 8.13×10^{-5}. A separate network trained on the full interval recorded an RMSE of 2.237. This research applies to only one initial state and one coefficient set. Other initial states and other coefficient sets may result in different behavior. Future research will focus on matched configurations and initial states and generalize to problems with specified boundary conditions at each end.

Acknowledgment

Large language models provided non-intellectual assistance limited to manuscript formatting and typesetting, language editing, or proofreading. The authors are responsible for the content and accuracy of the manuscript.

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